

The two principal boundary theorems extend to the full Schur class on every finite polydisk

Autonomous proof packet for arXiv:1002.3727

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Status. Candidate full answer for the two principal *function-theoretic* results, likely valid pending expert review. The result does not assert that a non-Schur-Agler function has a Hilbert-space model or an operator realization.

1 Source question and scope

Agler, McCarthy and Young ask at the end of *A Carathéodory theorem for the bidisk via Hilbert space methods*, arXiv:1002.3727, whether their results extend to functions in the Schur class, rather than only the Schur-Agler class, of the tridisk [1]. Their two principal function-theoretic theorems are Theorems 2.10 and 2.15 in the corrected arXiv version:

- (i) at a B-point, all inward directional derivatives exist and depend holomorphically and homogeneously on the direction, with a half-plane/Pick slope description;
- (ii) at a C-point, the ordinary gradient converges nontangentially to the angular gradient, and conversely at a B-point.

The question is visible in the source paper on PDF page 43:

Lying in a Koranyi region at τ is intermediate between approaching nontangentially and plurinontangentially. If $S = \{\lambda_t : 0 \leq t < 1\}$ then

S approaches τ nt $\Rightarrow S$ approaches τ in a Koranyi region $\Rightarrow S$ approaches τ pnt.

Jafari obtains results along similar lines to Abate. We should mention that the two authors also study holomorphic maps from a polydisk to another polydisk and to more general codomains.

Finally, we observe that we do not know whether our results extend to functions in the Schur class (rather than the Schur-Agler class) of the tridisk.

We prove both principal conclusions for the full Schur class $\mathcal{S}(\mathbb{D}^d)$, for every finite d . The directional theorem holds at every point of $\partial\mathbb{D}^d$, not only on the distinguished boundary $\mathbb{T}^d$. On $\mathbb{T}^d$ it has the natural higher-dimensional slope form used by Agler–Evans–Lykova–Young in their 2026 Schur–Agler theorem [4]; on a lower-dimensional face the slope ignores the coordinates lying in the open disk. When exactly two coordinates are unimodular, the slope has precisely the Pick-class parametrization in the source theorem. We do not claim model identities for functions outside $\mathcal{SA}_d$, since such functions need not have Agler models.

2 Definitions

Write $\|z\|_\infty = \max_j |z_j|$. A set $S \subset \mathbb{D}^d$ approaches $\tau \in \partial\mathbb{D}^d$ *nontangentially* if $\tau \in \overline{S}$ and there is $C < \infty$ such that

$$\|z - \tau\|_\infty \leq C(1 - \|z\|_\infty), \quad z \in S.$$

For $\varphi \in \mathcal{S}(\mathbb{D}^d)$, a point $\tau \in \partial\mathbb{D}^d$ is a *B-point* (or carapoint) if

$$\alpha := \liminf_{z \rightarrow \tau} \frac{1 - |\varphi(z)|}{1 - \|z\|_\infty} < \infty.$$

For $\tau \in \partial\mathbb{D}^d$, put

$$J(\tau) = \{j : |\tau_j| = 1\}, \quad J(\tau)^c = \{1, \dots, d\} \setminus J(\tau).$$

The cone of inward directions is

$$\mathcal{H}_\tau = \{\delta \in \mathbb{C}^d : \operatorname{Re}(\overline{\tau_j} \delta_j) > 0 \text{ for } j \in J(\tau)\}.$$

For $\delta \in \mathcal{H}_\tau$, $\tau - t\delta \in \mathbb{D}^d$ for all sufficiently small $t > 0$.

Write $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}$. The Pick class consists of the holomorphic functions on the upper half-plane with nonnegative imaginary part.

We say a holomorphic function f has an *angular differential* (ω, η) at τ if, on every nontangential set,

$$f(z) = \omega + \eta \cdot (z - \tau) + o(\|z - \tau\|_\infty).$$

For Schur functions, the source calls such a boundary point a C-point when $|\omega| = 1$.

3 Proof intuition

There are two independent mechanisms.

For directional derivatives, the full-Schur Julia inequality already controls the scaled quotients

$$q_t(\delta) = \frac{\varphi(\tau - t\delta) - \omega}{t}$$

locally uniformly on the inward cone. Thus they form a normal family. In every boundary coordinate, take $\delta_j = \tau_j a_j$, $a_j > 0$, and allow the remaining coordinates to range over a small complex polydisc. Each resulting straight ray is the radial image of a genuine analytic disk in $\mathbb{D}^d$. The classical one-variable Julia–Carathéodory theorem gives convergence there. This mixed totally-real/complex slice is a uniqueness set, so all normal-family cluster limits coincide everywhere. The limiting slope has nonpositive real part and is entire in every interior-face coordinate; Liouville’s theorem therefore makes it independent of those coordinates.

For C-points, surround every point z in a nontangential region by a polydisc of radius comparable to $1 - \|z\|_\infty$. The union remains in a slightly wider nontangential region. Cauchy’s estimate applied to the little- o affine remainder makes its gradient tend to zero. The converse follows by integrating the gradient along line segments to the boundary point.

4 Full-Schur directional derivatives

We use the full-Schur polydisk Julia inequality of Włodarczyk–Jafari–Abate [2, 1]: if τ is a B-point of a nonconstant $\varphi \in \mathcal{S}(\mathbb{D}^d)$, then $\alpha > 0$, there is $\omega \in \mathbb{T}$, and

$$\frac{|\varphi(z) - \omega|^2}{1 - |\varphi(z)|^2} \leq \alpha \max_{j \in J(\tau)} \frac{|z_j - \tau_j|^2}{1 - |z_j|^2}, \quad z \in \mathbb{D}^d, \quad (1)$$

The same theorem identifies ω as the nontangential boundary value and gives the radial identity

$$\lim_{r \uparrow 1} \frac{1 - |\varphi(r\tau)|}{1 - r} = \alpha. \quad (2)$$

Theorem 1 (Full-Schur directional theorem). *Let $d \geq 1$, let $\varphi \in \mathcal{S}(\mathbb{D}^d)$ be nonconstant, and let $\tau \in \partial\mathbb{D}^d$ be a B-point with Julia number α . Then there are $\omega \in \mathbb{T}$ and a holomorphic function $h : \mathcal{H}_\tau \rightarrow \mathbb{C}$ such that*

(i) $\operatorname{Re} h(\delta) < 0$ for every $\delta \in \mathcal{H}_\tau$;

(ii) $h(\tau) = -\alpha$;

(iii) for every $\delta \in \mathcal{H}_\tau$,

$$D_{-\delta}\varphi(\tau) := \lim_{t \downarrow 0} \frac{\varphi(\tau - t\delta) - \omega}{t} = \omega h(\delta);$$

(iv) h is homogeneous of degree one: $h(c\delta) = ch(\delta)$ whenever $\delta, c\delta \in \mathcal{H}_\tau$.

(v) h is independent of δ_k for every $k \notin J(\tau)$.

In particular, the directional derivative exists in every inward direction and is holomorphic and homogeneous in that direction. If $J(\tau) = \{j\}$, then

$$h(\delta) = -\alpha \overline{\tau_j} \delta_j.$$

If $J(\tau) = \{j, k\}$, there is a function p , analytic on $\mathbb{C} \setminus (-\infty, 0]$, such that p and $-zp(z)$ belong to the Pick class, p is real-valued on $(0, \infty)$, $p(1) = -\alpha$, and

$$h(\delta) = \overline{\tau_k} \delta_k p\left(\frac{\overline{\tau_k} \delta_k}{\overline{\tau_j} \delta_j}\right). \quad (3)$$

Thus, for two active coordinates, this recovers the exact Pick-function conclusion of Theorem 2.10 of the source. The theorem applies to all of $\mathcal{S}(\mathbb{D}^d)$, not merely $\mathcal{SA}_d$.

Proof. Let ω be supplied by (1). For $t > 0$, on the expanding domain

$$\Omega_t = \{\delta \in \mathcal{H}_\tau : \tau - t\delta \in \mathbb{D}^d\},$$

define

$$q_t(\delta) = \frac{\varphi(\tau - t\delta) - \omega}{t}.$$

We first prove local boundedness. If $K \Subset \mathcal{H}_\tau$, then for all sufficiently small t and $\delta \in K$,

$$\max_{j \in J(\tau)} \frac{|t\delta_j|^2}{1 - |\tau_j - t\delta_j|^2} = t \max_{j \in J(\tau)} \frac{|\delta_j|^2}{2 \operatorname{Re}(\overline{\tau_j} \delta_j) - t|\delta_j|^2} \leq C_K t.$$

Set $x = |\varphi(\tau - t\delta) - \omega|$ and $s = |\varphi(\tau - t\delta)|$. From (1),

$$x^2 \leq \alpha C_K t(1 - s^2) \leq 2\alpha C_K t(1 - s) \leq 2\alpha C_K tx.$$

Consequently $x \leq 2\alpha C_K t$, so the q_t are locally bounded. Montel's theorem and a diagonal exhaustion imply that every sequence $t_n \downarrow 0$ has a subsequence for which q_{t_n} converges locally uniformly on $\mathcal{H}_\tau$ to a holomorphic function.

It remains to prove uniqueness of this cluster function. Write $J = J(\tau)$. Let $a_j > 0$ for $j \in J$, put $A = \max_{j \in J} a_j$, and choose $\xi_k \in \mathbb{C}$, $k \notin J$, so that

$$\frac{2|\xi_k|}{A} < 1 - |\tau_k|.$$

Define $\delta_j^a = \tau_j a_j$ for $j \in J$ and $\delta_k^a = \xi_k$ for $k \notin J$. Define the analytic disk by

$$\Psi_a(\zeta)_j = \tau_j(1 - c_j + c_j \zeta), \quad c_j = a_j/A, \quad j \in J,$$

and

$$\Psi_a(\zeta)_k = \tau_k + \frac{\xi_k}{A}(\zeta - 1), \quad k \notin J, \quad \zeta \in \mathbb{D}.$$

Since $0 < c_j \leq 1$ in the active coordinates and the displayed size condition handles the other coordinates, $\Psi_a(\mathbb{D}) \subset \mathbb{D}^d$. Put $G_a = \varphi \circ \Psi_a$. For $0 < r < 1$, inequality (1) gives

$$\frac{|G_a(r) - \omega|^2}{1 - |G_a(r)|^2} \leq \alpha \max_{j \in J} \frac{c_j^2(1 - r)^2}{1 - (1 - c_j + c_j r)^2} = \alpha \max_j \frac{c_j(1 - r)}{2 - c_j(1 - r)} \leq \alpha(1 - r).$$

The preceding elementary estimate then gives $|G_a(r) - \omega| = O(1 - r)$. Thus 1 is a B-point of the disk Schur function G_a . The classical Julia–Carathéodory theorem implies that

$$\lim_{r \uparrow 1} \frac{G_a(r) - \omega}{r - 1}$$

exists. One has $\Psi_a(1 - At) = \tau - t\delta^a$, and hence $\lim_{t \downarrow 0} q_t(\delta^a)$ exists.

In particular, take $1 < a_j < 2$ for $j \in J$ and let each ξ_k range in a sufficiently small fixed disk about zero. Any two holomorphic cluster functions agree on this product of a real open box in the active coordinates and a complex open polydisc in the inactive coordinates. It is a uniqueness set: first iterate the one-variable identity theorem through the active coordinates, and then through the inactive coordinates. All cluster functions are consequently equal. Normality now implies that the whole family q_t converges locally uniformly to a holomorphic function q on $\mathcal{H}_\tau$.

For every real $c > 0$, direct rescaling gives $q(c\delta) = cq(\delta)$. For fixed δ , both sides are holomorphic in every complex c for which $c\delta \in \mathcal{H}_\tau$. This set of c 's is convex and contains the positive real axis, so the one-variable identity theorem gives complex homogeneity.

Set $h = \bar{\omega}q$. Since $\varphi(\tau - t\delta) = \omega + tq(\delta) + o(t)$ and $|\varphi| \leq 1$,

$$\operatorname{Re} h(\delta) = \operatorname{Re}(\bar{\omega}q(\delta)) \leq 0.$$

For the radial direction, apply the one-variable theorem to $r \mapsto \varphi(r\tau)$ and use (2); this gives $q(\tau) = -\omega\alpha$, hence $h(\tau) = -\alpha$. The harmonic maximum principle on the connected cone $\mathcal{H}_\tau$ upgrades $\operatorname{Re} h \leq 0$ to $\operatorname{Re} h < 0$.

Fix the active coordinates of δ . As a function of any inactive coordinate, h is entire and has real part bounded above by zero. The function $\exp h$ is bounded entire, so Liouville's theorem shows

that h is constant in that coordinate. This proves independence of all inactive coordinates. If there is only one active coordinate j , homogeneity and $h(\tau) = -\alpha$ now give $h(\delta) = -\alpha\bar{\tau}_j\delta_j$.

Suppose finally that $J = \{j, k\}$. Put $a = \bar{\tau}_j\delta_j$, $b = \bar{\tau}_k\delta_k$, and write $F(a, b) = h(\delta)$; the inactive coordinates do not matter. Homogeneity defines an analytic function p on $\mathbb{C} \setminus (-\infty, 0]$ by

$$F(a, b) = bp(b/a), \quad a, b \in \mathbb{H}.$$

For positive a, b , the disk Julia–Carathéodory theorem used above says that $F(a, b)$ is real and negative, so p is real-valued on $(0, \infty)$, and $p(1) = F(1, 1) = -\alpha$. If z lies in the upper half-plane, choose $a \in \mathbb{H}$ with $az \in \mathbb{H}$. Since

$$\operatorname{Re}\{azp(z)\} = \operatorname{Re} F(a, az) < 0,$$

letting the argument of a tend to the two endpoints of its permitted interval gives respectively

$$\operatorname{Im} p(z) \geq 0, \quad \operatorname{Im}(-zp(z)) \geq 0.$$

Thus both p and $-zp(z)$ are Pick functions, and (3) follows. All assertions are proved. $\square$

Remark 2 (Relation to known normal-family arguments). McCarthy and Pascoe prove directional differentiability by a normal-family/uniqueness argument in a noncommutative setting [3]. Their scalar Corollary 6.2 assumes $\|\psi\|_{H^\infty(G_\gamma)} = 1$, which on $\mathbb{D}^d$, $d \geq 3$, is the Schur–Agler (multiplier) condition, not merely $\|\psi\|_\infty \leq 1$. The proof above replaces their model-derived estimate by the older Julia inequality (1), which is valid for the full Schur class on every polydisk.

5 C-points and ordinary gradients

The next theorem is independent of Schur-class theory.

Theorem 3 (Nontangential differential–gradient equivalence). *Let $f : \mathbb{D}^d \rightarrow \mathbb{C}$ be holomorphic and $\tau \in \partial\mathbb{D}^d$.*

(a) *If f has an angular differential (ω, η) at τ , then*

$$\nabla f(z) \longrightarrow \eta \quad \text{as } z \rightarrow \tau \text{ nontangentially.}$$

(b) *Conversely, if $f(z) \rightarrow \omega$ and $\nabla f(z) \rightarrow \eta$ nontangentially, then f has the angular differential (ω, η) at τ .*

Consequently, for every $\varphi \in \mathcal{S}(\mathbb{D}^d)$ and every B-point τ , τ is a C-point if and only if the nontangential limit of $\nabla\varphi$ exists.

Proof. For (a), fix a nontangential set S of aperture C , and write

$$f(z) = \omega + \eta \cdot (z - \tau) + e(z).$$

Choose $0 < a < 1$. For $\lambda \in S$, put $d_\lambda = 1 - \|\lambda\|_\infty$ and let

$$P_\lambda = \{z : \|z - \lambda\|_\infty < ad_\lambda\}.$$

Every P_λ lies in $\mathbb{D}^d$. Moreover, for $z \in P_\lambda$,

$$1 - \|z\|_\infty \geq (1 - a)d_\lambda, \quad \|z - \tau\|_\infty \leq (C + a)d_\lambda.$$

Thus the union of these polydiscs, for λ near τ , is itself a nontangential set. The angular differential on that union implies

$$\sup_{z \in P_\lambda} |e(z)| = o(d_\lambda).$$

Cauchy's estimate on the coordinate circle of radius $ad_\lambda/2$ now yields

$$|\partial_j e(\lambda)| \leq \frac{2}{ad_\lambda} \sup_{z \in P_\lambda} |e(z)| = o(1).$$

Hence $\nabla f(\lambda) \rightarrow \eta$ on S .

For (b), fix a nontangential set S , and adjoin the open line segment from τ to every point of S :

$$V_S = \{\tau + s(\lambda - \tau) : \lambda \in S, 0 < s \leq 1\}.$$

The polydisk is convex, and

$$1 - \|\tau + s(\lambda - \tau)\|_\infty \geq s(1 - \|\lambda\|_\infty), \quad \|\tau + s(\lambda - \tau) - \tau\|_\infty = s\|\lambda - \tau\|_\infty.$$

So V_S is nontangential. Integrating on the segment and using the assumed boundary limits gives

$$f(\lambda) - \omega - \eta \cdot (\lambda - \tau) = \int_0^1 (\nabla f(\tau + s(\lambda - \tau)) - \eta) \cdot (\lambda - \tau) ds.$$

The supremum of the gradient error over these segments tends to zero as $\lambda \rightarrow \tau$ in S , so the right side is $o(\|\lambda - \tau\|_\infty)$.

Finally, a Schur B-point has a unimodular nontangential function limit by the polydisk Julia theorem. Applying (a) and (b) gives the stated C-point equivalence. $\square$

Corollary 4 (Higher-order angular jets). *Suppose $m \geq 1$, P_m is a polynomial of degree at most m , and a holomorphic $f : \mathbb{D}^d \rightarrow \mathbb{C}$ satisfies*

$$f(z) = P_m(z - \tau) + o(\|z - \tau\|_\infty^m)$$

on every nontangential set at τ . Then, for every multi-index β with $|\beta| \leq m$,

$$\partial^\beta f(z) - \partial^\beta P_m(z - \tau) = o(\|z - \tau\|_\infty^{m-|\beta|})$$

nontangentially.

Proof. Repeat the scaled-polydisc construction in Theorem 3. On a fixed nontangential region, $1 - \|z\|_\infty$ and $\|z - \tau\|_\infty$ are comparable. The multivariable Cauchy estimate applied to $f - P_m$ gives a factor $(1 - \|z\|_\infty)^{-|\beta|}$, and the claimed estimate follows. $\square$

6 What is and is not answered

Theorem 1 proves the full d -variable version of the first principal boundary conclusion at every boundary face: existence, holomorphic dependence, degree-one homogeneity, half-plane sign, radial normalization, independence of inactive coordinates, the exact one-active-coordinate formula, and the source's Pick parametrization when two coordinates are active. Theorem 3 proves the second principal theorem and its converse in every dimension, in fact for arbitrary holomorphic functions once the boundary value is assumed. In particular both statements hold on the tridisk for Schur functions that are not Schur-Agler.

The packet does *not* claim that the model-theoretic characterizations in [1, 4] extend to all Schur functions. Nor does it give an operator formula for h outside $\mathcal{SA}_d$. For three or more active coordinates, the two-variable Pick formula is replaced by the natural homogeneous left-half-plane slope on the active inward cone. For exactly two active coordinates, formula (3) recovers the source formula verbatim.

7 Verification and novelty status

The proof was checked line by line against the exact full-Schur Julia inequality and radial identity in Abate, Lemma 3.1 and Theorem 3.1 [2]. The normal-family mechanism was compared with McCarthy–Pascoe, arXiv:1606.09629, Section 5 and Corollary 6.2 [3]. Their scalar result has the stronger multiplier/Schur–Agler norm hypothesis. The current arXiv:2508.13742v2 (17 July 2026) still states its higher-dimensional directional theorem only for $\mathcal{SA}_d$ and only on $\mathbb{T}^d$ [4].

A bounded search on 17 August 2026 used the exact source-question sentence, “directional derivative”/“carapoint”/“angular gradient” with polydisk/tridisk, the source title and authors, and citations to arXiv:1002.3727. No explicit full-Schur theorem matching Theorems 1–3 was found. Because the directional proof is a short synthesis of two known mechanisms, novelty confidence is *moderate*; mathematical-confidence is *high pending specialist review*.

The most important review points are:

1. confirm that the full-Schur Julia inequality (1) is used with exactly the source B-point normalization;
2. audit the normal-family uniqueness step on the mixed totally-real/complex slice and the Liouville reduction on lower-dimensional faces;
3. decide whether the source’s broad phrase “our results” should be read as its two principal function-theoretic theorems (answered here) or as also demanding model/realization statements (explicitly outside scope).

References

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- [3] J. E. McCarthy and J. E. Pascoe, *A non-commutative Julia inequality*, Math. Ann. 370 (2018), 423–446; arXiv:1606.09629.
- [4] J. Agler, C. Evans, Z. A. Lykova and N. J. Young, *Boundary behavior of functions in the Schur–Agler class of the polydisc*, arXiv:2508.13742v2, 17 July 2026.